# AHMAD HAZWAN SYAUQEE BIN HUSAIRI

Computer Science Graduate | Software Engineering | AI, Computer Vision, IoT & Data

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## SUMMARY

Computer Science (Intelligent Computing) graduate from USM with 6 months of industry experience at **MIMOS Berhad**, where I extended a legacy PHP/MySQL inventory system with new modules, role-based access and audit trails. I build across the stack, from training a **YOLOv8** road hazard model that won Silver at PIXEL 2026 to deploying the web apps around it on Google Cloud Run. Also built IoT prototypes, data dashboards and a Flutter app. Open to software engineering, AI, IoT and data roles.

## EDUCATION

**Bachelor of Computer Science (Intelligent Computing) with Honours** | *Universiti Sains Malaysia, Penang* 2022 – 2026

- Final CGPA: 3.30 | Final semester GPA: 3.65
- Relevant coursework: Machine Learning, IoT Architecture & Smart Applications, Business Intelligence & Analytics, Big Data Technologies, Parallel & Cloud Computing, Web Engineering

**Foundation in Engineering** | *Pusat Asasi UiTM Dengkil, Selangor* 2021 – 2022

- GPA: 4.00 | MUET: Band 4.5

**Sijil Pelajaran Malaysia (SPM)** | *SMK Seksyen 4 Kota Damansara, Selangor* 2016 – 2020

- Results: 5A 4B

## PROFESSIONAL EXPERIENCE

**Industrial Intern** | *MIMOS Berhad* Mar 2025 – Sep 2025

- Extended a legacy PHP/MySQL inventory system (DCIMS) by designing and building two new modules from scratch, End User Computing and Software Asset Management, including new database schemas, LDAP login, audit trails and assignment history tracking.
- Prototyped workflows in **Figma** and built the front-end for device listings, dashboards, and data entry forms with JavaScript/jQuery and Bootstrap, adding real-time feedback via toast notifications and a recent-activity dashboard.
- Implemented role-based access control across five user tiers (Server, Network, EUC, Software, and Super Admin) and restructured device categorisation from a location-based to a category-based system, improving lookup and usability across 650+ device records.
- Wrote UAT scripts and led acceptance testing with four admin groups, fixing permission, validation and usability issues.
- Deployed the system from local XAMPP to a live test server via WinSCP, fixing permission and configuration issues.
- Wrote the user manual and system architecture diagram, working with senior engineers to align features with user needs.

## TECHNICAL PROJECTS

**Road Hazard Detection System** | *Final Year Project* | *GitHub* · *Demo video* 2025 – 2026

- Trained a **YOLOv8** model for potholes and cracks on **8,560 labelled Malaysian road images**, reaching **79.8% mAP@0.5** on validation.
- Built tropical augmentation (rain, canopy shade, glare, haze) that generates extra labelled training samples and stress-tests predictions across six road conditions.
- Built a mobile-first React web app for citizens, plus an admin dashboard and contractor portal, on a **Flask API deployed to Google Cloud Run** with Firebase Firestore and Storage.
- Built **human-in-the-loop retraining**: admin corrections feed new training data, promoted only if mAP improves.

**Sightline: Valorant Heatmap** | *Side project* | *Live demo* 2026

- Pulled **1,358 ranked matches** from the HenrikDev API and processed them with **pandas, NumPy and SciPy**, mapping every player position onto minimap coordinates.
- Built interactive heatmaps and a round timing chart, filterable by map, rank, side, agent and round phase; deployed on Netlify.

**Product Catalog App** | *Flutter* | *GitHub* · *Walkthrough video* 2026

- Built a **Flutter** app on the DummyJSON REST API with infinite scroll, a detail screen and debounced search.
- Modelled screen state as a **sealed class** (loading, empty, error with retry, success) and split the code into data and presentation layers, with a repository handling pagination.
- Added pull-to-refresh, image loading and error placeholders, and **unit tests** for JSON parsing.

**Smart Drainage Pipe Monitor** | *IoT team project* | *Demo video* 2025

- Built a smart-city IoT prototype using an **ESP32** microcontroller with ultrasonic and flow sensors to detect drain blockages.
- Integrated hardware telemetry with a cloud backend via **HTTP REST** to Firebase Realtime Database, feeding a live web dashboard and on-site LED and buzzer alerts.

**Superstore Performance Dashboard** | *Business Intelligence coursework* 2026

- Built an interactive **Tableau** dashboard on the Superstore retail dataset to expose the gap between revenue and net margin across categories, regions and time.
- Found a high-volume, low-margin pattern in Furniture using bar charts, dual-axis trend lines and treemaps, and turned it into three business recommendations.

**Toxland VR: Hop & Help** | *Animation & VR Coursework* 2026

- Migrated an existing VR board game to Unity 6.3 for the **PICO 4** headset, resolving compatibility issues across the upgrade.
- Produced the accompanying project report and presentation materials.

## LEADERSHIP & ACTIVITIES

**Exco, Multimedia & Publicity** | *Desasiswa Aman Damai, USM* | *Designs* 2022 – 2024

- Ran the college's official social media and event publicity, delivering designs through leadership approval before each deadline.
- As acting unit head for inter-college sports (Sukad), scheduled exco coverage across simultaneous games and led poster design.

**Silver Award, PIXEL 2026** | *School of Computer Sciences, Universiti Sains Malaysia* 2026

- Awarded Silver for the Road Hazard Detection System at PIXEL 2026, USM's School of Computer Sciences project showcase.

**Entrant, IEM IDEAS Competition 2026** | *Institution of Engineers Malaysia (IEM), Individual Category* 2026

- Entered the Road Hazard Detection System into IEM's Innovation, Design, and Engineering for Advanced Sustainability (IDEAS) Competition.

## SKILLS

**Programming:** Python, PHP, JavaScript, SQL, Java, C++, Dart, R, HTML/CSS

**Web:** React.js, Flask, Node.js / Express, Bootstrap, jQuery, Vite, Leaflet.js

**Databases:** MySQL, PostgreSQL (Supabase), Firebase Firestore / Realtime Database, Oracle APEX

**AI & Data:** YOLOv8, data augmentation, pandas, NumPy, SciPy, Tableau, Altair AI Studio, sentiment analysis

**Cloud & DevOps:** Google Cloud Platform (Cloud Run), Firebase, Docker (basic), Git & GitHub

**IoT:** ESP32, Arduino, ultrasonic and flow sensors, MQTT

**Mobile:** Flutter, Provider, Android Studio / Emulator

**Game & VR:** Unity (6.3), PICO 4 VR development

**Design & Productivity:** Figma, Canva, Blender, Adobe Photoshop, Microsoft Office, VS Code

**Spoken Languages:** Malay (native), English (fluent)

## CERTIFICATIONS

- Certificate of Industrial Internship, MIMOS Berhad | Sept 2025 (24-week residency in full-lifecycle software engineering and legacy modernisation)
- Data-Driven Decision Making: AIoT and V-ONE Data Analytics Perspective, ViTrox Academy | Dec 2025

## REFERENCE

**Dr. Nurul Hashimah** | Final Year Project Supervisor, USM | nurulhashimah@usm.my | +604 653 4645